

CO2- AT2 (Application based Questions)

10 Use case diagram - Smart hotel Reservation system

Analysis

a) Actor

- Guest
- Manager
- Payment service (external)
- admin (Generalized user)

b) Use cases

- Register/login
- Search rooms
- Check availability
- Make reservation
- Cancel reservation
- Make payment
- View booking history
- Allocate rooms
- Generate reports

c) relationships

«Include» Search rooms → check availability

«Include» make reservation → make payment

«extend» cancel reservation, refund payment

Generalization: user → guest, manager

Use case diagram

Smart hotel reservation system

User

/ \

Guest manager

| |

Search room
reservation
cancel book
booking list

Allocate room
generate report

|

Include

|

check availability

|

«Include»

|

make payment ← payment service.

|

«extend»

|

refund payment

2. class diagram - university management system

Classes

a) Student

→ student ID

→ name

→ enroll(course)

b) faculty

→ faculty ID

→ name

→ teach(course)

c) course

→ course ID

→ title

→ assign(faculty)

d) department

→ dept ID

→ dept name

→ manage(course)

e) examination

→ exam ID

→ exam date

→ conduct exam()

f) relationship

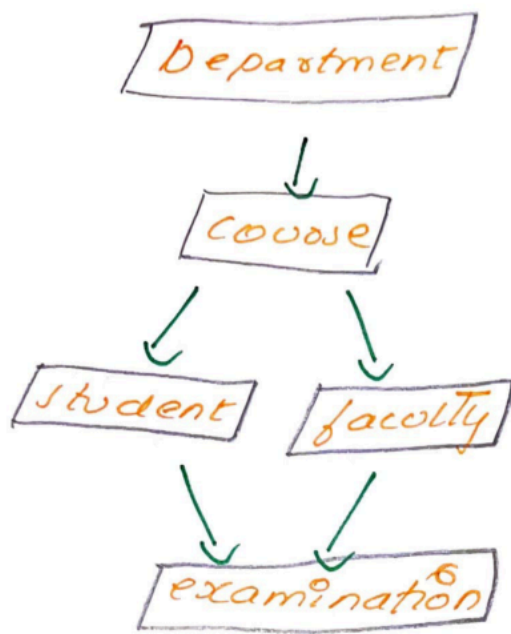
→ Department - course

→ faculty - course

→ student - course

→ course - examination

class diagram



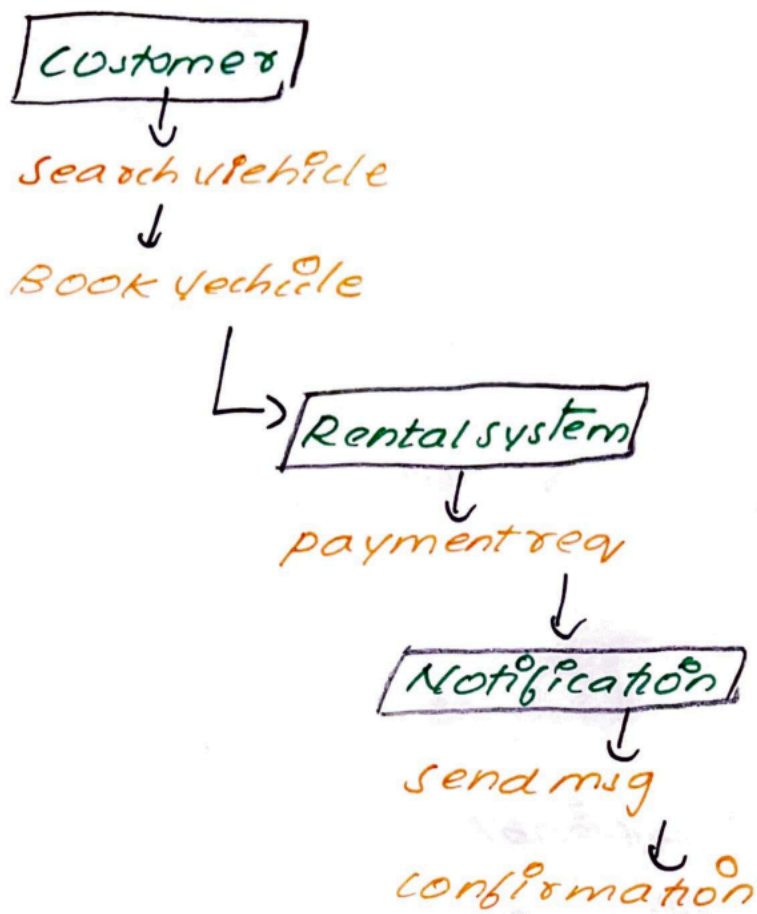
30

Sequence diagram - vehicle rental system

Objects.

- Customer
- rental system
- Vehicle Database
- payment gateway
- notification service.

Sequence diagram:

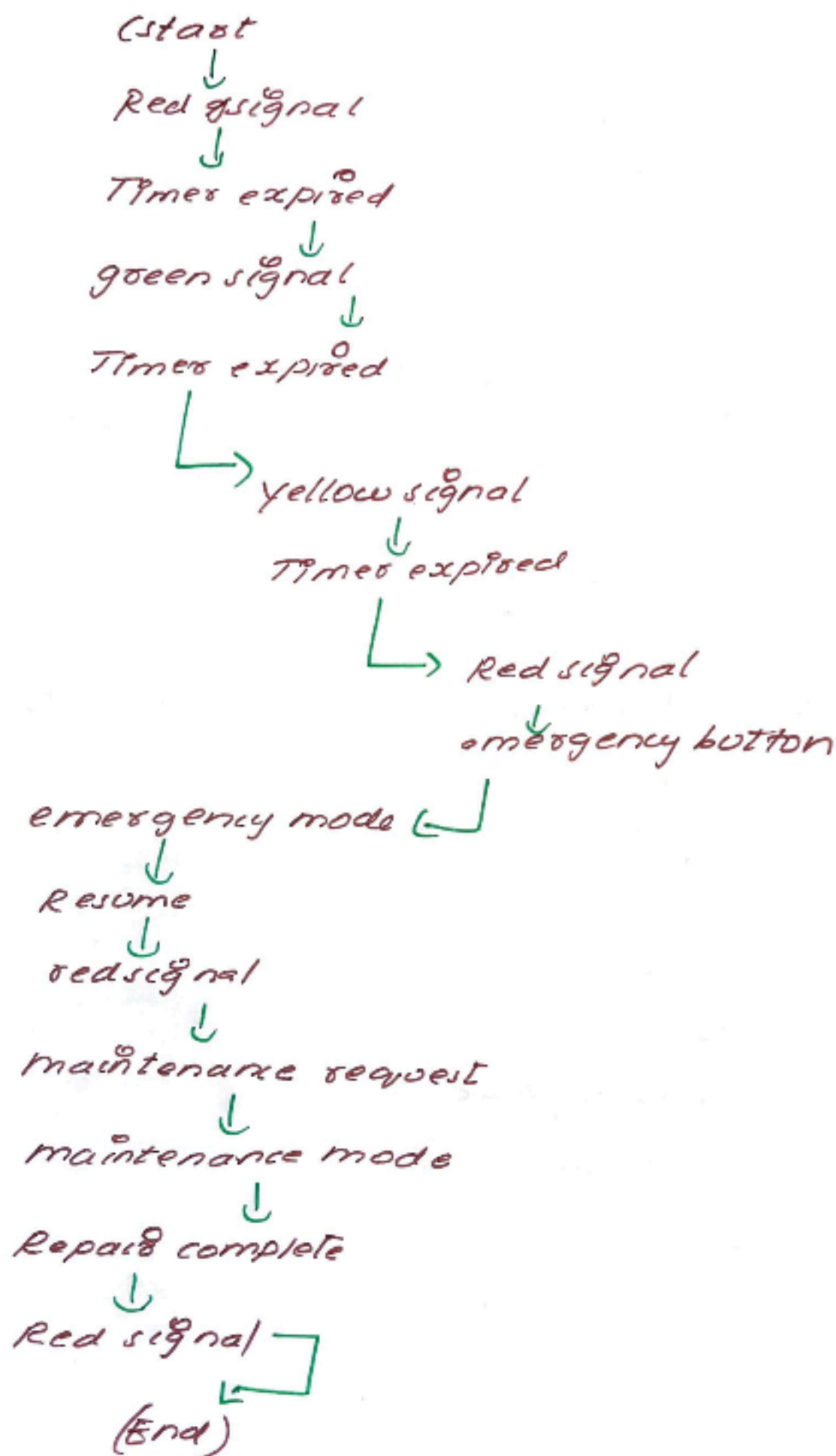


4. state diagram - Traffic signal control system

a) states

- Initial
- Red
- green
- yellow
- emergency mode
- maintenance mode
- final

state diagram:

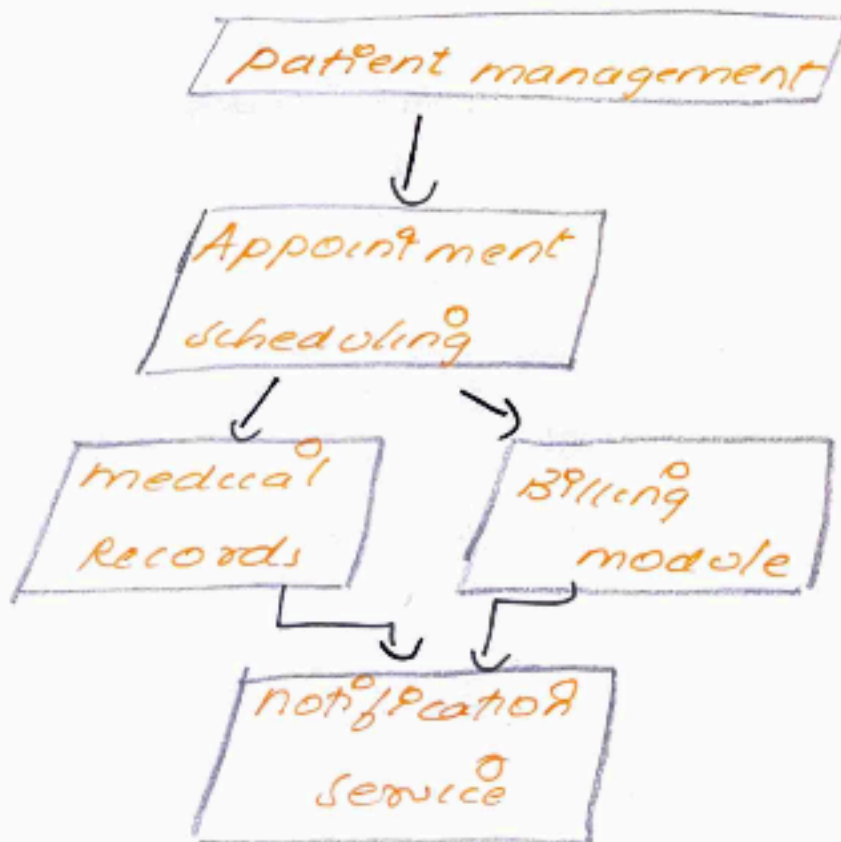


5. Component diagram - smart healthcare portal

a) Components

- patient management
- appointment scheduling
- medical records
- Billing module
- notification service.

Dependencies:



6. Deployment diagram - smart city waste management system

hardware nodes

- smart waste bin
- edge device
- cloud server
- database server
- monitoring dashboard

Deployment diagram:

